

STEAM AND HOT SURFACES TOOLBOX TALK

Steam can be difficult to see, and even a small leak can cause severe burns.

PURPOSE AND SCOPE

Prevent thermal burns and pressurised releases from steam, condensate, hot piping and process equipment.

KEY HAZARDS

1  Direct contact with high-temperature steam.	2  Fine and difficult-to-see steam jets.	3  Discharge of hot condensate.	4  Uninsulated piping and equipment.
5  Trapped pressure inside steam lines.	6  Incorrect or premature valve operation.	7  Slip hazards from wet surfaces.	8  Inadequate gloves or clothing for thermal protection.

CRITICAL POINTS TO VERIFY IN THE FIELD

01 Direct contact with high-temperature steam.	02 Fine and difficult-to-see steam jets.	03 Discharge of hot condensate.
---	---	--



REMEMBER — SERNEM

S

Stop and think

E

Evaluate the risk

R

Remove or reduce

N

Never assume

E

Ensure protection

M

Monitor and communicate

STEAM AND HOT SURFACES TOOLBOX TALK

Steam can be difficult to see, and even a small leak can cause severe burns.

FIELD REQUIREMENTS AND VERIFICATION

Prevent thermal burns and pressurised releases from steam, condensate, hot piping and process equipment.

SERNEM FIELD 01

FIELD VERIFICATION 1

Hot lines identified

SERNEM FIELD 02

FIELD VERIFICATION 2

Insulation checked

SERNEM FIELD 03

FIELD VERIFICATION 3

Leak method suitable

SERNEM FIELD 04

FIELD VERIFICATION 4

Pressure verified

VERIFICATION BEFORE START

- | | |
|---|---|
| ☐ Hot lines identified | ☐ Insulation checked |
| ☐ Leak method suitable | ☐ Pressure verified |
| ☐ Temperature verified | ☐ Drainage safe |

SUPERVISOR DISCUSSION POINTS

- Where are the hot surfaces?
- Is any insulation missing?
- How will steam leaks be checked?

FIELD APPLICATION



Steam can be difficult to see, and even a small leak can cause severe burns.



STEAM AND HOT SURFACES TOOLBOX TALK

Steam can be difficult to see, and even a small leak can cause severe burns.

CRITICAL CONTROL MEASURES

1. ELIMINATE / REDUCE THE RISK

Identify hot piping and equipment before work. • Route condensate drains to a safe location. • Remove slip hazards created by condensate.

2. ENGINEERING & PHYSICAL CONTROLS

Barricade or guard areas with missing insulation. • Open steam valves slowly and under control. • Restore insulation and protective coverings after maintenance.

3. WORK METHOD & FIELD CONTROLS

Use an approved method rather than hands to detect steam leaks. • Wear suitable thermal gloves, face protection and clothing.

4. PPE, VERIFICATION & SUPERVISION

Verify pressure and temperature before opening systems. • Immediately isolate areas containing steam leaks.

CRITICAL WARNING



Team, a small steam leak can still cause a major injury. We will not use our hands to locate leaks, and we will verify pressure, temperature and insulation before starting work.

SITE CHECKLIST — BEFORE STARTING WORK

- | | |
|--|---|
| ☐ Hot lines identified | ☐ Insulation checked |
| ☐ Leak method suitable | ☐ Pressure verified |
| ☐ Temperature verified | ☐ Drainage safe |
| ☐ Valve method understood | ☐ PPE suitable |
| ☐ Area barricaded | ☐ Crew briefed |



STEAM AND HOT SURFACES TOOLBOX TALK

Steam can be difficult to see, and even a small leak can cause severe burns.

ATTENDANCE AND APPROVAL

Project / Site: _____ Date: _____

Presented by: _____ Work area: _____

No	Full Name	Company / Role	Signature
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			